Intelligent VPN Traffic Analysis and Classification Framework on Raspberry Pi Using Machine Learning Techniques

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***Abstract*—The increasing adoption of Virtual Private Networks (VPNs) for secure communications has created new challenges regarding traffic monitoring, classification, and anomaly detection in resource-constrained environments. This paper presents an intelligent framework for VPN traffic analysis and classification deployed on a Raspberry Pi platform. The proposed system combines secure VPN connectivity using OpenVPN and WireGuard with real-time traffic monitoring and machine learning (ML)-based analysis. Network traffic is collected and processed using packet capture and inspection tools, while Random Forest (RF) and Support Vector Machine (SVM) models are employed to classify traffic flows and identify potentially suspicious activities. Furthermore, TensorFlow Lite is integrated to enable lightweight real-time inference on embedded hardware with limited computational resources. Experimental evaluation demonstrates the feasibility of deploying intelligent traffic analysis services directly on edge devices, providing a cost-effective solution for secure Small Office/Home Office (SOHO) environments. The obtained results indicate that ML techniques can effectively enhance traffic visibility and support cybersecurity monitoring while maintaining low resource consumption on Raspberry Pi-based platforms.**

Keywords*—VPN, Raspberry Pi, traffic analysis, traffic classification, machine learning, Random Forest, Support Vector Machine, TensorFlow Lite, edge computing, cybersecurity.

I. Introduction

The rapid growth of remote connectivity services, cloud-based applications, and distributed computing environments has significantly increased the demand for secure communication mechanisms [1], [2]. VPNs have become one of the most widely adopted technologies for protecting data confidentiality, ensuring secure remote access, and safeguarding sensitive information transmitted over public networks [3]. VPN solutions such as OpenVPN and WireGuard provide encrypted communication channels that reduce the risk of eavesdropping, unauthorized access, and traffic manipulation [4].

At the same time, the increasing volume and diversity of network traffic have introduced new challenges for network administrators and cybersecurity professionals [5]. Although VPN technologies successfully protect data confidentiality, they may also obscure traffic visibility, making the identification of network anomalies, malicious activities, and traffic patterns more difficult [6]. Traditional traffic monitoring approaches often rely on manual inspection or signature-based detection mechanisms, which may be insufficient when dealing with large-scale and continuously evolving network environments [7].

Recent advances in ML have demonstrated significant potential for improving network traffic analysis and cybersecurity monitoring [8]. Supervised learning algorithms such as RF and SVM have been successfully employed for traffic classification, anomaly detection, and intrusion identification tasks [9], [10]. Furthermore, lightweight artificial intelligence frameworks enable the deployment of intelligent monitoring services directly on embedded and edge computing platforms, reducing latency and minimizing dependence on cloud-based infrastructures [2].

In parallel, edge computing has emerged as a promising paradigm for processing data closer to its source [11]. Single-board computers such as Raspberry Pi provide a cost-effective and energy-efficient platform capable of supporting networking services, local data processing, and ML inference [12]. Their low power consumption and flexibility make them attractive candidates for SOHO environments, where affordable cybersecurity solutions are increasingly required [1].

Despite the growing interest in VPN technologies and ML-based traffic analysis, relatively few studies have investigated the integration of secure VPN communication, real-time traffic monitoring, and intelligent traffic classification within a unified edge-computing framework. Most existing approaches focus either on VPN deployment or on traffic classification separately, without addressing the challenges associated with resource-constrained embedded platforms [4].

To address this research gap, this paper proposes an intelligent VPN traffic analysis and classification framework deployed on a Raspberry Pi platform. The proposed system integrates OpenVPN and WireGuard for secure communication, network traffic monitoring tools for packet acquisition, and ML techniques for traffic classification and analysis. RF and SVM models are employed to identify and categorize network traffic, while TensorFlow Lite is utilized to support lightweight real-time inference suitable for embedded devices. The proposed framework aims to provide enhanced traffic visibility, improved cybersecurity monitoring capabilities, and efficient operation in resource-constrained environments.

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The main contribution of this work consists in the development of an intelligent VPN traffic analysis and classification framework deployed on a Raspberry Pi edge-computing platform. The proposed system combines secure communication services provided through OpenVPN and WireGuard with real-time traffic monitoring and ML-based analysis mechanisms. Furthermore, RF and SVM models are integrated to perform network traffic classification, while TensorFlow Lite enables lightweight inference suitable for resource-constrained embedded environments. The framework was experimentally validated in a SOHO-oriented scenario, demonstrating the feasibility of combining VPN services, traffic monitoring, and intelligent analysis within a unified low-cost platform. The obtained results highlight the potential of edge-based ML techniques for enhancing traffic visibility, supporting cybersecurity monitoring, and enabling efficient network management in modern distributed environments.

The remainder of this paper is organized as follows. Section II presents the related work and background regarding VPN technologies, edge computing platforms, and ML-based traffic analysis. Section III describes the proposed framework and system architecture. Section IV details the experimental setup and implementation methodology. Section V presents the obtained results and performance evaluation. Finally, Section VI concludes the paper and outlines future research directions.

II. Related Work

The rapid evolution of cybersecurity threats, encrypted communications, and edge computing technologies has stimulated extensive research efforts in the fields of network traffic analysis, intrusion detection, and intelligent monitoring systems. Recent studies have investigated various approaches based on ML, DL, and edge intelligence to improve network visibility and support cybersecurity operations. Nevertheless, the integration of secure VPN communication, intelligent traffic classification, and lightweight deployment on embedded platforms remains an active research area.

Razooqi and Pekar [13] proposed a wavelet-based ML framework for binary VPN traffic detection, focusing on distinguishing VPN-encrypted traffic from non-VPN communications regardless of the underlying application. The authors evaluated RF, Neural Networks (NN), and SVM using wavelet-derived temporal features extracted from network flows. Experimental results demonstrated that RF achieved the best performance, reaching an F1-score of 99% and maintaining 98% accuracy even after significant dataset filtering, while NN achieved an F1-score of 98% and SVM obtained a maximum F1-score of 90%. Furthermore, the study showed that deeper wavelet decomposition levels improved classification performance, although the gains were relatively small compared to the additional computational overhead. However, the proposed approach focused exclusively on binary VPN detection and did not investigate real-time traffic monitoring, VPN service deployment, or lightweight edge implementation on resource-constrained platforms such as Raspberry Pi.

Elmaghraby et al. [14] proposed a ML framework for encrypted network traffic classification, aiming to identify different application categories, including web browsing, Voice over Internet Protocol (VoIP), file transfer, and video streaming, without decrypting packet contents. The authors investigated three architectures combining NN, Bidirectional Long Short-Term Memory (LSTM), RF, SVM, k-Nearest Neighbors (k-NN), and ensemble voting techniques. Experimental results showed that the first approach achieved the highest classification accuracy of 96.8%, while the second and third approaches achieved 95.2% and 96.5%, respectively. Furthermore, individual classifiers such as RF and k-NN reached accuracies of 99.6% and 100% in specific configurations, demonstrating the effectiveness of hybrid ML models for encrypted traffic analysis. However, the study focused on application-level traffic classification and did not address VPN deployment, real-time network monitoring, or edge-based implementation on resource-constrained platforms such as Raspberry Pi.

Sattar et al. [15] proposed Encrypted Traffic Anomaly Detection using Self-Supervised Contrastive Learning (ET-SSL), a privacy-preserving framework for anomaly detection in encrypted network traffic. The method leverages self-supervised contrastive learning to extract discriminative representations from flow-level features, including packet lengths, inter-arrival times, flow durations, and protocol metadata, without requiring packet decryption or labeled datasets. Experimental evaluation on the Canadian Institute for Cybersecurity (CIC)-Darknet2020, Information Security Centre of Excellence (ISCX) VPN-nonVPN, and University of New South Wales (UNSW)-NB15 datasets demonstrated classification accuracies of 96.8%, 94.3%, and 95.1%, respectively, with false positive rates as low as 1.2%. The proposed approach also achieved a 92.7% true positive rate for zero-day attack detection, processing up to 10 Gbps of encrypted traffic with a latency of 15-25 ms, making it suitable for real-time deployment. Furthermore, the framework outperformed conventional RF, K-Means, and Autoencoder-based approaches in terms of accuracy, F1-score, and robustness against previously unseen attacks. However, the study focused on anomaly detection in encrypted traffic and did not investigate VPN deployment architectures, network management strategies, or embedded implementations dedicated to secure remote access systems.

Wijethilaka et al. [16] proposed a cost-effective ML-based Network Intrusion Detection System (NIDS) deployed on a Raspberry Pi 4 for real-time network traffic analysis. The framework combines multiple ML algorithms, including RF, LSTM, Artificial NN (ANN), XGBoost, and Naïve Bayes, to detect and classify a wide range of cyberattacks such as Denial of Service (DoS), Distributed DoS (DDoS), Structured Query Language (SQL) Injection, Port Scanning, Malware, Reconnaissance, Man-in-the-Middle (MitM), Brute Force, Ransomware, Worms, and Exploits. To address dataset imbalance, the authors employed the Synthetic Minority Oversampling Technique (SMOTE), significantly improving detection performance for minority attack classes. The system was implemented on a low-cost Raspberry Pi 4 platform and integrated with real-time alerting mechanisms including Light-Emitting Diodes (LEDs), buzzers, and email notifications. Experimental evaluation on the NetFlow-University of Queensland-NIDS (NF-UQ-NIDS) dataset demonstrated a maximum detection accuracy of 96.5%, a precision of 94.1%, a recall of 95.5%, and an F1-score of 94.8% using the RF classifier. Furthermore, the proposed NIDS achieved an average response time of approximately 50 ms, outperforming several traditional intrusion detection systems while maintaining low hardware costs. Although the solution provides an effective and affordable approach for

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real-time network security monitoring, the study focuses primarily on intrusion detection and does not address VPN deployment, remote access architectures, or secure network tunneling mechanisms.

Wang and Gu [17] proposed Parameter-Efficient Fine-Tuning-based Encrypted Traffic Representation Learning Method (PETReLM), a DL framework designed for encrypted traffic classification in both network management and cybersecurity scenarios. The proposed approach combines a Bidirectional Encoder Representations from Transformers (BERT)-based pre-trained model with a Parameter-Efficient Fine-Tuning (PEFT) strategy using Low-Rank Adaptation (LoRA), significantly reducing the number of trainable parameters while maintaining high classification performance. The framework processes encrypted traffic at packet level, employing Byte-Pair Encoding and WordPiece tokenization to transform packet payloads into sequences suitable for transformer-based learning. The model is pre-trained using Masked Language Modeling (MLM) and a modified Next Sentence Prediction (mNSP) task, enabling the extraction of generalized encrypted traffic representations. Experimental validation was conducted on multiple public datasets including ISCXVPN2016, ISCXTor2017, University of Science and Technology of China - Traffic Flow Classification (USTC-TFC2016), CIC-IDS2017, and CIC Internet of Things (IoT) Dataset 2023. Results demonstrated classification accuracies of up to 99.98% for Tor service classification, 98.67% for VPN application classification, and 98.81% for malware traffic classification. Furthermore, the proposed framework reduced the trainable parameter count to approximately 5.6% of the Encrypted Traffic (ET)-BERT model while decreasing training time by 16.68% and reducing Graphics Processing Unit (GPU) memory consumption. The study highlights the potential of transformer-based architectures combined with parameter-efficient adaptation techniques for scalable encrypted traffic analysis in modern cybersecurity environments.

III. Proposed Framework

A. System Architecture

The proposed framework (shown in Fig. 1) was designed to provide secure communication, real-time traffic monitoring, and intelligent traffic classification within a unified edge-computing environment. The system combines VPN technologies, traffic analysis mechanisms, and ML algorithms deployed on a Raspberry Pi platform. The primary objective is to enhance network visibility and cybersecurity monitoring while maintaining low hardware costs and reduced computational requirements suitable for SOHO environments.

The overall architecture consists of five main components: the VPN communication layer, the traffic monitoring module, the feature extraction module, the ML classification engine, and the result visualization module. The VPN communication layer establishes secure communication channels using OpenVPN and WireGuard protocols, ensuring confidentiality and integrity of transmitted data. Network traffic generated through these secure tunnels is continuously monitored and collected by the traffic monitoring module, which captures packet-level and flow-level information required for further analysis.

The collected traffic data are subsequently processed by the feature extraction module, where relevant statistical and temporal characteristics are derived from the captured network flows. These features include packet sizes, packet counts, flow duration, inter-arrival times, and protocol-specific information. The extracted features are then provided to the ML engine, which employs RF and SVM classifiers to identify traffic patterns and support intelligent network analysis.

To enable efficient execution on resource-constrained hardware, the trained ML models are optimized and deployed using TensorFlow Lite. This approach reduces memory consumption and computational overhead while preserving classification performance. Finally, the generated results are displayed through a monitoring interface, allowing administrators to observe traffic behavior and classification outcomes in real time.

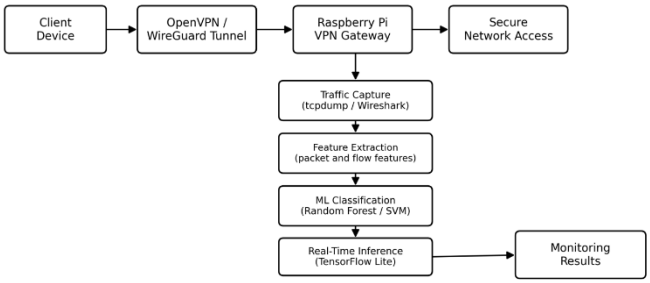


Fig. 1. Overall architecture of the proposed VPN traffic analysis and classification framework.

The proposed architecture enables the integration of secure VPN communication, traffic monitoring, and ML-based analysis within a single embedded platform. Unlike conventional solutions that treat these functions independently, the proposed framework provides a unified and cost-effective approach for intelligent network monitoring and cybersecurity support at the network edge.

B. VPN Communication Layer

The VPN communication layer represents the security foundation of the proposed framework, presented in Fig. 2, providing encrypted communication channels between remote clients and the Raspberry Pi edge platform. To ensure secure data transmission and support different operational requirements, two widely adopted VPN technologies, namely OpenVPN and WireGuard, were integrated into the system architecture.

OpenVPN was selected due to its maturity, extensive protocol support, and strong security mechanisms based on Secure Sockets Layer/Transport Layer Security (SSL/TLS) encryption. The protocol enables the establishment of secure tunnels over public networks while supporting authentication, encryption, and integrity verification services. Furthermore, its widespread adoption in enterprise and remote-access environments makes it a suitable candidate for secure communication infrastructures [18].

In parallel, WireGuard was incorporated as a lightweight alternative designed to provide reduced computational overhead and improved communication efficiency. Unlike traditional VPN solutions, WireGuard employs a simplified architecture and a modern cryptographic suite, enabling faster connection establishment and lower latency. These characteristics make it particularly attractive for embedded and edge-computing environments where computational resources are limited [19].

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Within the proposed framework, remote clients establish encrypted tunnels with the Raspberry Pi VPN gateway through either OpenVPN or WireGuard. All transmitted traffic is routed through the secure communication channel before reaching the protected network environment. This approach ensures the confidentiality and integrity of exchanged information while simultaneously generating encrypted traffic suitable for monitoring and classification purposes.

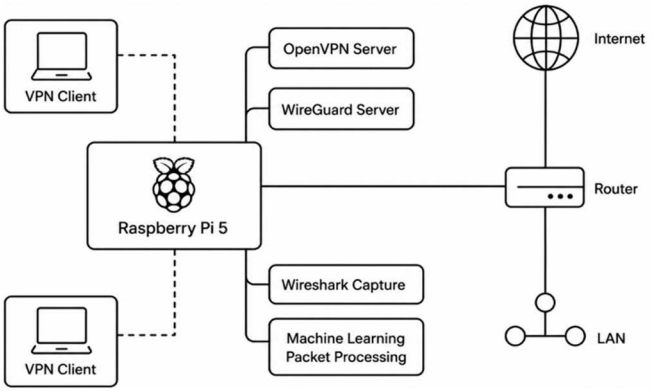


Fig. 2. VPN communication workflow implemented on the Raspberry Pi platform.

The integration of both VPN technologies enables the evaluation of traffic characteristics under different encryption mechanisms while maintaining a secure communication environment. The generated traffic serves as the primary input for the monitoring and ML modules described in the following sections.

C. Traffic Monitoring and Data Collection

The traffic monitoring module is responsible for capturing and observing the network packets generated through the VPN communication layer. After the VPN tunnel is established, the Raspberry Pi operates as an intermediate edge node that routes the client traffic and enables local packet inspection without interrupting the secure communication process. This configuration allows the system to collect relevant traffic information while preserving the encrypted nature of the transmitted data.

Traffic acquisition was performed using packet capture tools such as Wireshark (Fig. 3) and command-line monitoring utilities. These tools enable the visualization and extraction of packet-level information, including source and destination addresses, transport protocol, packet length, timestamps, and communication direction. Such information is essential for constructing the feature set used by the subsequent ML classification stage.

The collected traffic data are not analyzed based on packet payload inspection, since VPN communication encrypts the transmitted content. Instead, the proposed framework relies on metadata and statistical characteristics extracted from the observed traffic flows. This approach is suitable for encrypted environments, where direct content inspection is either technically impractical or undesirable from a privacy perspective.

The monitoring process therefore provides the connection between the VPN infrastructure and the intelligent classification module. By transforming raw packet observations into structured traffic descriptors, the system enables ML algorithms to identify traffic patterns, distinguish communication behaviors, and support real-time cybersecurity monitoring on the Raspberry Pi platform.

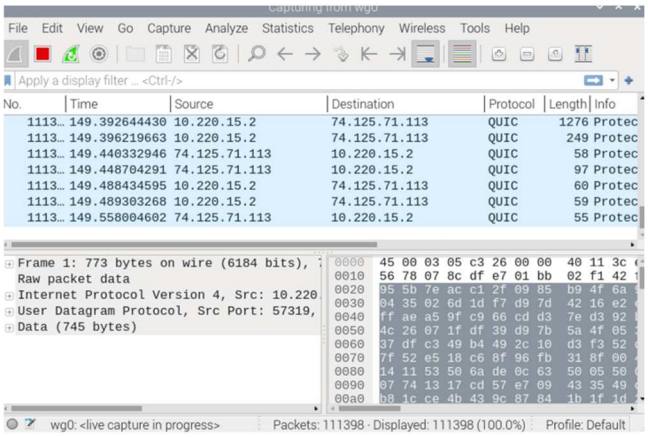


Fig. 3. Captured VPN traffic visualization using Wireshark during the monitoring stage.

D. Feature Extraction and Traffic Representation

The effectiveness of ML algorithms largely depends on the quality and relevance of the extracted traffic features. Since VPN technologies encrypt the packet payload, direct content inspection is not feasible. Consequently, the proposed framework relies on statistical and flow-based characteristics that can be obtained from captured network traffic without violating the confidentiality provided by the VPN tunnel.

After packet acquisition, the monitoring module processes the captured traffic and extracts a set of descriptive features from each communication flow. These features are selected to characterize the behavior of network sessions while remaining independent of the encrypted packet contents. The extracted attributes include packet length, flow duration, packet count, transmitted bytes, source and destination ports, communication protocol, and packet inter-arrival times.

The extracted parameters are subsequently organized into feature vectors that provide a compact numerical representation of each observed traffic flow. This representation enables the ML models to identify similarities and differences between traffic patterns generated by various network activities. Furthermore, the use of flow-based features reduces computational complexity and facilitates deployment on resource-constrained embedded platforms such as Raspberry Pi, as shown in Fig. 4.

Before being supplied to the classification stage, the extracted features undergo preprocessing operations to improve data quality and ensure consistent model behavior. These operations include data normalization, removal of incomplete records, and formatting procedures required by the ML framework. The resulting feature vectors constitute the input data used by the RF and SVM classifiers described in the following section.

The adopted feature extraction strategy enables efficient analysis of encrypted traffic while preserving user privacy. By focusing exclusively on traffic metadata and statistical characteristics, the proposed framework remains applicable to both OpenVPN and WireGuard communication environments without requiring access to the encrypted payload contents.

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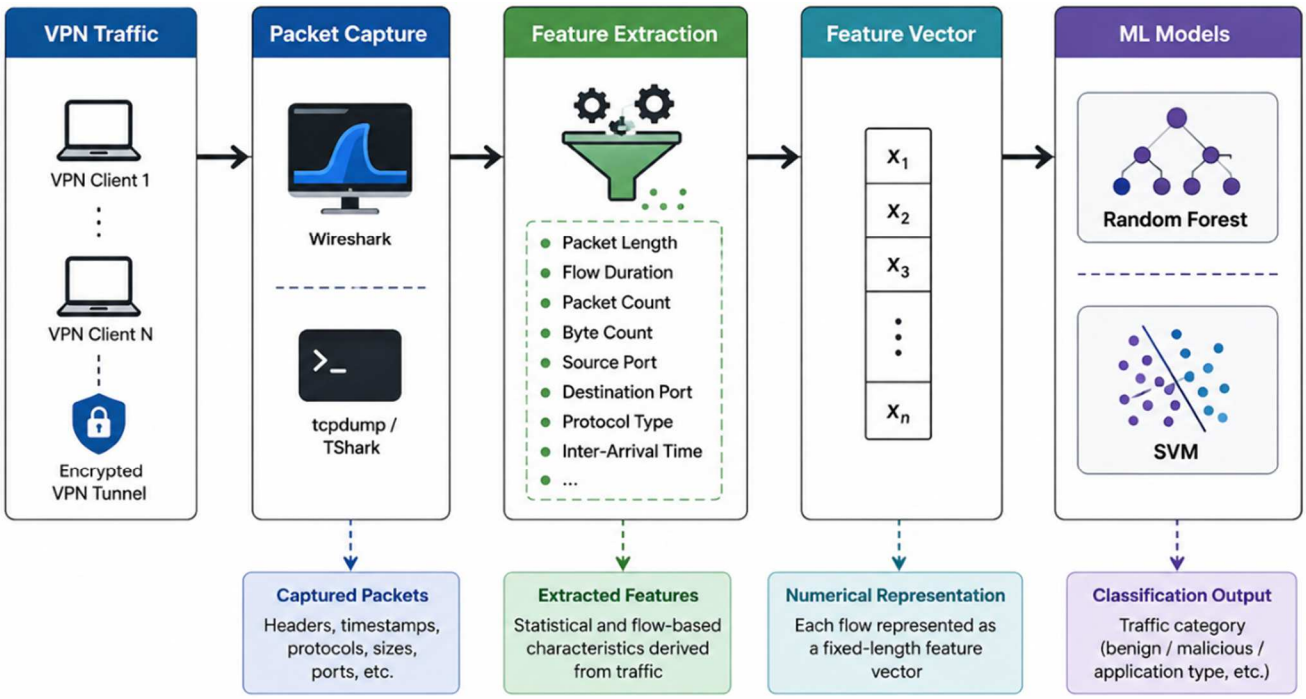


Fig. 4. Feature extraction process used for traffic representation and ML analysis.

E. ML-Based Traffic Classification

As illustrated in Fig. 5, the extracted traffic features were analyzed using supervised ML algorithms to identify traffic patterns and support intelligent network monitoring. Among the numerous classification techniques available in the literature, RF and SVM were selected due to their proven effectiveness in network traffic analysis, robustness against noisy data, and suitability for deployment on resource-constrained edge platforms.

RF is an ensemble learning method that combines multiple decision trees to improve classification accuracy and reduce overfitting. During the training process, each decision tree is constructed using a randomly selected subset of features and training samples. The final classification result is obtained through majority voting among all generated trees. This approach enables the model to capture complex relationships within the traffic data while maintaining high generalization capability [20].

The RF classifier was employed as an ensemble learning technique for network traffic classification. The model was configured using 200 decision trees, allowing the classifier to improve robustness and reduce overfitting through the aggregation of multiple independent decision-making processes. Prior to training, feature selection was performed using correlation analysis, retaining the most relevant traffic attributes with respect to the target protocol class. The resulting feature vectors were standardized and subsequently used for model training and evaluation. The adoption of Random Forest was motivated by its ability to handle high-dimensional traffic data, provide strong generalization performance, and maintain stable classification accuracy in complex network environments.

SVM represents another widely adopted classification technique for network security applications. The algorithm attempts to identify an optimal decision boundary that maximizes the separation margin between different traffic classes. By projecting the feature vectors into a higher-dimensional space, SVM can effectively distinguish between traffic categories that are not linearly separable in the original feature space. This characteristic makes the algorithm particularly suitable for traffic classification tasks involving encrypted communications [21].

The SVM classifier was implemented using a Radial Basis Function (RBF) kernel to model non-linear relationships within the extracted traffic features. To improve classification performance, hyperparameter optimization was performed through GridSearchCV, exploring different values of the regularization parameter C and kernel coefficient gamma. Furthermore, stratified sampling was applied during dataset partitioning to preserve class distributions across training and testing subsets. The SVM algorithm was selected due to its strong capability to identify optimal decision boundaries in high-dimensional feature spaces and its proven effectiveness in network traffic classification tasks.

Both classifiers receive the feature vectors generated during the extraction stage and produce traffic classification outputs based on the learned patterns. The classification results can be used to distinguish between different communication behaviors, identify traffic categories, and support real-time network monitoring. Furthermore, the simultaneous evaluation of RF and SVM allows a comparative analysis of their performance within the proposed VPN traffic analysis framework [22].

The adoption of these ML techniques enables the proposed system to perform intelligent traffic analysis without requiring access to encrypted packet contents. Consequently, the framework preserves the privacy benefits provided by VPN technologies while still maintaining visibility into network behavior through metadata-based analysis.

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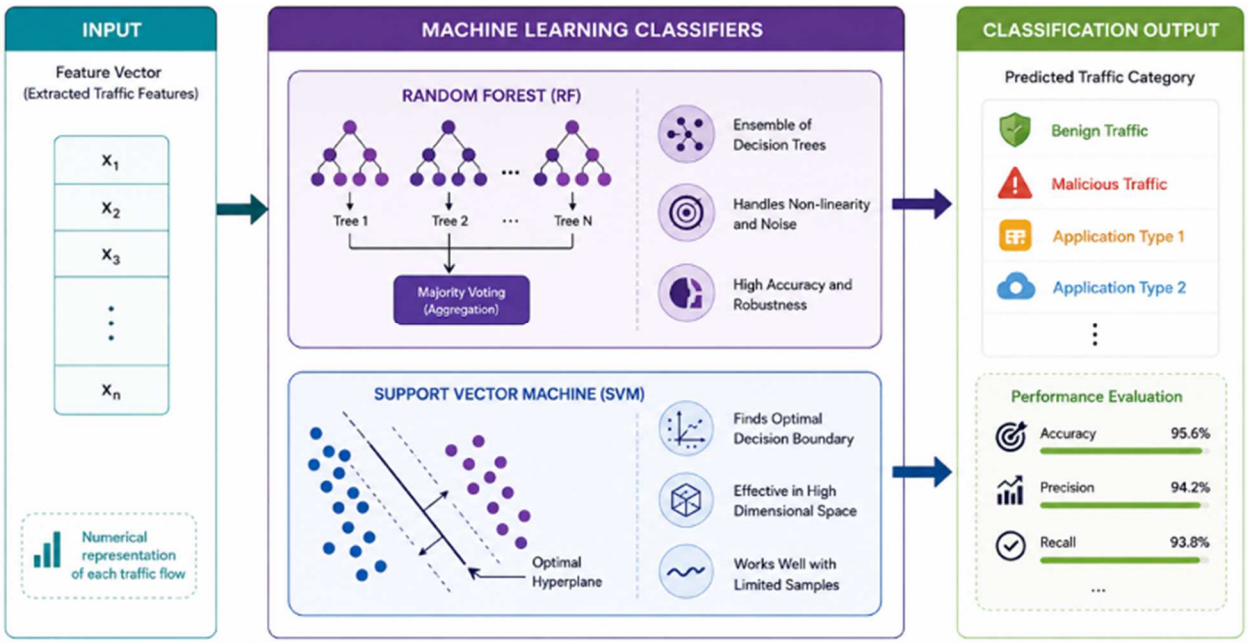


Fig. 5. ML-based traffic classification process using RF and SVM models.

F. TensorFlow Lite Edge Deployment

To enable efficient execution on resource-constrained hardware, the proposed framework (Fig. 6) incorporates TensorFlow Lite as the inference engine for ML deployment on the Raspberry Pi platform. TensorFlow Lite is specifically designed for embedded and edge-computing applications, providing reduced memory consumption, lower computational requirements, and improved inference performance compared to conventional ML deployment frameworks [23].

After the training phase, the developed ML models are converted into lightweight formats suitable for execution on the Raspberry Pi device. This optimization process reduces model size while preserving the classification capabilities required for real-time traffic analysis. As a result, the proposed framework can perform intelligent traffic classification locally without relying on external cloud services or remote processing infrastructure.

The deployment architecture enables the Raspberry Pi to simultaneously perform VPN services, traffic monitoring, feature extraction, and ML inference within a single embedded platform. This integration minimizes communication latency and improves system responsiveness, making the solution suitable for real-time cybersecurity monitoring applications.

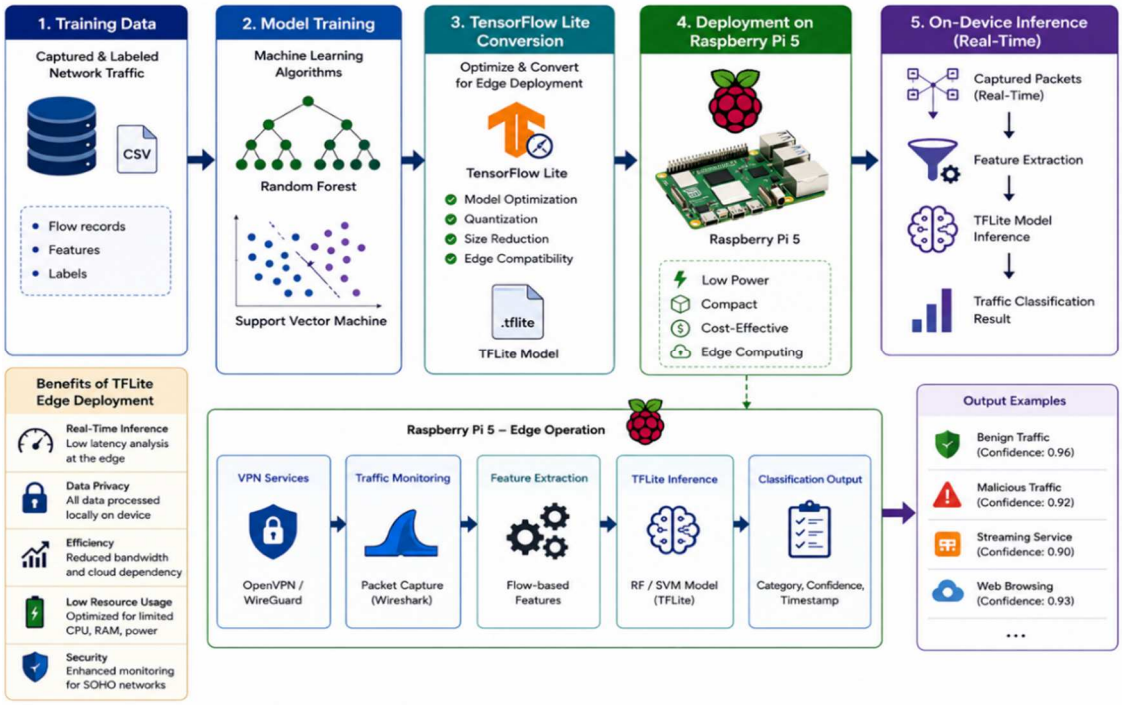


Fig. 6. TensorFlow Lite deployment workflow on the Raspberry Pi platform.

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By executing the inference process locally, the framework also improves data privacy, since captured traffic information remains within the monitored environment. Furthermore, local processing reduces network overhead and eliminates the dependency on continuous Internet connectivity for classification tasks.

To support real-time inference on the Raspberry Pi platform, a lightweight neural network was implemented using TensorFlow and subsequently converted to TensorFlow Lite format. The architecture consisted of two fully connected hidden layers containing 128 and 64 neurons respectively, employing Rectified Linear Unit (ReLU) activation functions and a dropout rate of 0.3 to mitigate overfitting. Model training was performed using the Adam optimizer and categorical cross-entropy loss function over ten training epochs. Following the training stage, the model was converted to TensorFlow Lite with optimization and float16 quantization enabled, reducing memory requirements and improving execution efficiency on the embedded platform.

The adoption of TensorFlow Lite demonstrates the feasibility of combining VPN communication services, network traffic monitoring, and intelligent ML-based analysis on a low-cost embedded platform. Consequently, the proposed framework provides a practical solution for SOHO environments that require secure communication, traffic visibility, and lightweight cybersecurity capabilities.

IV. Experimental Results and Discussion

A. Experimental Setup

The experimental evaluation was conducted using a Raspberry Pi 5 platform configured as an edge-computing node capable of providing VPN services, traffic monitoring functionality, and ML-based traffic analysis. The hardware platform was selected due to its low cost, low power consumption, and suitability for SOHO cybersecurity deployments. The experimental environment integrated OpenVPN and WireGuard services, allowing encrypted traffic to be generated and monitored under realistic operating conditions. The Raspberry Pi platform simultaneously performed packet acquisition, feature extraction, traffic classification, and real-time inference tasks. The overall experimental configuration employed throughout the evaluation process is illustrated in Fig. 7.

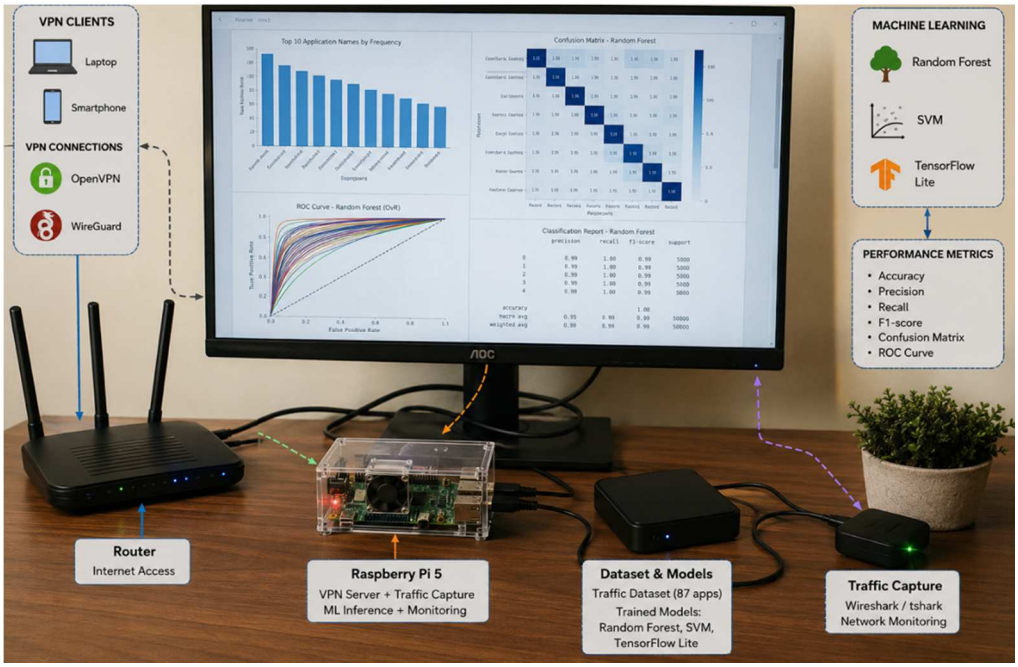


Fig. 7. Experimental setup used for VPN traffic acquisition, ML model training, and edge-based inference on the Raspberry Pi platform.

The software environment consisted of Raspberry Pi Operating System (OS) (64-bit), OpenVPN 2.6, WireGuard 1.0, Wireshark/tshark 4.2, Python 3.11, Scikit-learn 1.5, TensorFlow 2.16, and TensorFlow Lite 2.16. Traffic monitoring was performed directly on the VPN interfaces, enabling the collection of network flow information from encrypted communication sessions. Captured traffic data were subsequently exported and processed using dedicated ML pipelines developed for the proposed framework.

The experimental platform was based on a Raspberry Pi 5 equipped with a quad-core 64-bit Advanced RISC (Reduced Instruction Set Computer) Machines (Arm) Cortex-A76 processor operating at 2.4 GHz, 2 GB Low Power Double Data Rate (LPDDR4X) memory, Gigabit Ethernet connectivity, and Raspberry Pi OS [24]. The device served as the central node for VPN communication, traffic monitoring, ML inference, and real-time network analysis.

Representative measurements obtained during live operation indicated a CPU utilization of approximately 21.6%, memory utilization of 35.1%, and a processor temperature of about 40°C, while maintaining network throughput levels of 2.26 Mbps (transmit) and 2.37 Mbps (receive). No stability issues were observed during the experimental campaign, confirming the practical feasibility of deploying intelligent VPN traffic monitoring and classification services on low-cost edge-computing platforms intended for SOHO cybersecurity applications.

For the ML experiments, the publicly available Internet Protocol (IP) Network Traffic Flows Labeled with 87 Applications dataset was employed [25]. This dataset contains

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labeled network traffic flows associated with many applications and communication services, making it suitable for evaluating traffic classification algorithms under realistic conditions. During the experimental phase, approximately 50,000 traffic flow records were processed and analyzed for training and testing purposes.

Prior to model training, the dataset underwent a preprocessing stage that included the removal of incomplete records, elimination of irrelevant attributes, label encoding of protocol classes, and feature normalization using the StandardScaler method. Additionally, correlation-based feature selection was applied to identify the most informative traffic characteristics and reduce data dimensionality while preserving classification performance. This procedure contributed to improving model efficiency and reducing computational overhead during the training and inference stages.

The dataset was divided into training and testing subsets using an 80%-20% partitioning strategy. For the SVM implementation, stratified sampling was employed to preserve class distributions across both subsets and ensure a balanced evaluation process. The RF classifier was configured using 200 decision trees, while the SVM model employed a RBF kernel optimized through a GridSearchCV procedure. In parallel, a lightweight neural network architecture was implemented using TensorFlow, consisting of two hidden layers with 128 and 64 neurons, respectively, followed by conversion into TensorFlow Lite format for deployment on the Raspberry Pi platform.

Model performance was evaluated using widely adopted classification metrics, including Accuracy, Precision, Recall, and F1-score. Furthermore, confusion matrices and detailed classification reports were generated to provide a comprehensive assessment of the traffic classification capabilities of the implemented models. The experimental evaluation also considered the practical feasibility of deploying ML-based traffic analysis on a low-cost embedded platform, focusing on the ability to perform real-time inference while maintaining acceptable computational resource utilization.

The adopted experimental setup provides a realistic environment for validating the proposed intelligent VPN traffic analysis framework. By combining secure VPN communication, traffic monitoring, ML-based classification, and edge deployment within a unified Raspberry Pi platform, the experimental configuration enables a comprehensive evaluation of both classification performance and practical deployment feasibility in real-world networking scenarios.

B. Performance Evaluation Metrics

To assess the effectiveness of the proposed VPN traffic classification framework, several widely adopted ML evaluation metrics were employed. These metrics provide a comprehensive assessment of the classification performance and enable a detailed comparison between the implemented RF, SVM, and TensorFlow Lite-based models.

The primary performance indicator used in this study is Accuracy, which represents the proportion of correctly classified traffic instances relative to the total number of analyzed samples. Accuracy provides an overall measure of classification effectiveness [26] and is defined as (1):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where TP denotes true positives, TN true negatives, FP false positives, and FN false negatives [26].

Although accuracy provides a global assessment of model performance, it may not fully reflect the classifier behavior when dealing with multiple traffic classes. Therefore, Precision was also considered. Precision measures the proportion of correctly identified positive instances among all samples classified as positive [26] (2):

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

A high precision value indicates that the classifier produces a low number of false alarms and incorrectly classified traffic flows. Another important metric is Recall, which quantifies the ability of the classifier to correctly identify all relevant traffic instances belonging to a specific class [26] (3):

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

Recall is particularly relevant in cybersecurity applications, where the failure to identify a specific traffic category may reduce the effectiveness of the monitoring system. To provide a balanced evaluation that simultaneously considers both Precision and Recall, the F1-score was employed. This metric represents the harmonic mean between Precision and Recall and is defined as [26] (4):

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{4}$$

The F1-score is especially useful when comparing classifiers operating on datasets containing multiple traffic categories and varying class distributions.

In addition to these numerical metrics, confusion matrices and detailed classification reports were generated for each evaluated model. These tools provide class-level performance information and facilitate the identification of traffic categories that may be more difficult to distinguish. The combination of Accuracy, Precision, Recall, F1-score, and confusion matrix analysis provide a comprehensive framework for evaluating the effectiveness of the proposed intelligent VPN traffic classification system.

C. Classification Results

The proposed ML framework was evaluated using three classification approaches, namely RF, SVM, and TensorFlow Lite-based neural network inference. The experiments were conducted using the preprocessed network traffic dataset described in the previous section, while the evaluation was performed using Accuracy, Precision, Recall, and F1-score metrics.

Prior to model training, an exploratory analysis of the dataset was performed to investigate the distribution of traffic classes and the characteristics of the extracted network flows. The employed dataset contains approximately 50,000 traffic

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records described by 87 traffic-related attributes and multiple application categories, including Hypertext Transfer Protocol (HTTP), SSL, GOOGLE, AMAZON, YOUTUBE, MICROSOFT, and WINDOWS_UPDATE traffic. The distribution of the analyzed traffic classes is presented in Fig. 8.

[5 rows x 87 columns]
Number of Rows: 50000
Number of Columns: 87

	ProtocolName
0	HTTP_PROXY
1	HTTP
2	HTTP_CONNECT
3	SSL
4	GOOGLE
5	YOUTUBE
6	FACEBOOK
7	CONTENT_FLASH
8	DROPBOX
9	WINDOWS_UPDATE
10	AMAZON
11	MICROSOFT
12	TOR
13	GMAIL
14	YAHOO
15	MSN
16	SSL_NO_CERT
17	SKYPE
18	MS_ONE_DRIVE
19	MSSQL
20	TWITTER

Fig. 8. Overview of the network traffic dataset containing 50,000 traffic flows and multiple application classes used for training and evaluation.

To improve classification efficiency and reduce computational complexity, a feature-selection stage was applied. Correlation analysis was initially performed to identify dependencies among the extracted traffic characteristics. Subsequently, the most relevant 29 features were selected and retained for model training. The resulting feature correlation matrix is illustrated in Fig. 9.

The RF classifier achieved strong classification performance across all evaluated traffic categories. The initial evaluation resulted in an overall classification accuracy of approximately 92.33%, while most application classes obtained F1-score values exceeding 0.90. As illustrated in Fig. 10, the classifier demonstrated particularly good performance for HTTP, MICROSOFT, SSL, and WINDOWS_UPDATE traffic classes.

The confusion matrix associated with the Random Forest model is presented in Fig. 11. Most samples are concentrated along the main diagonal, indicating successful classification of most traffic flows. Only a limited number of misclassifications were observed between traffic categories exhibiting similar communication characteristics.

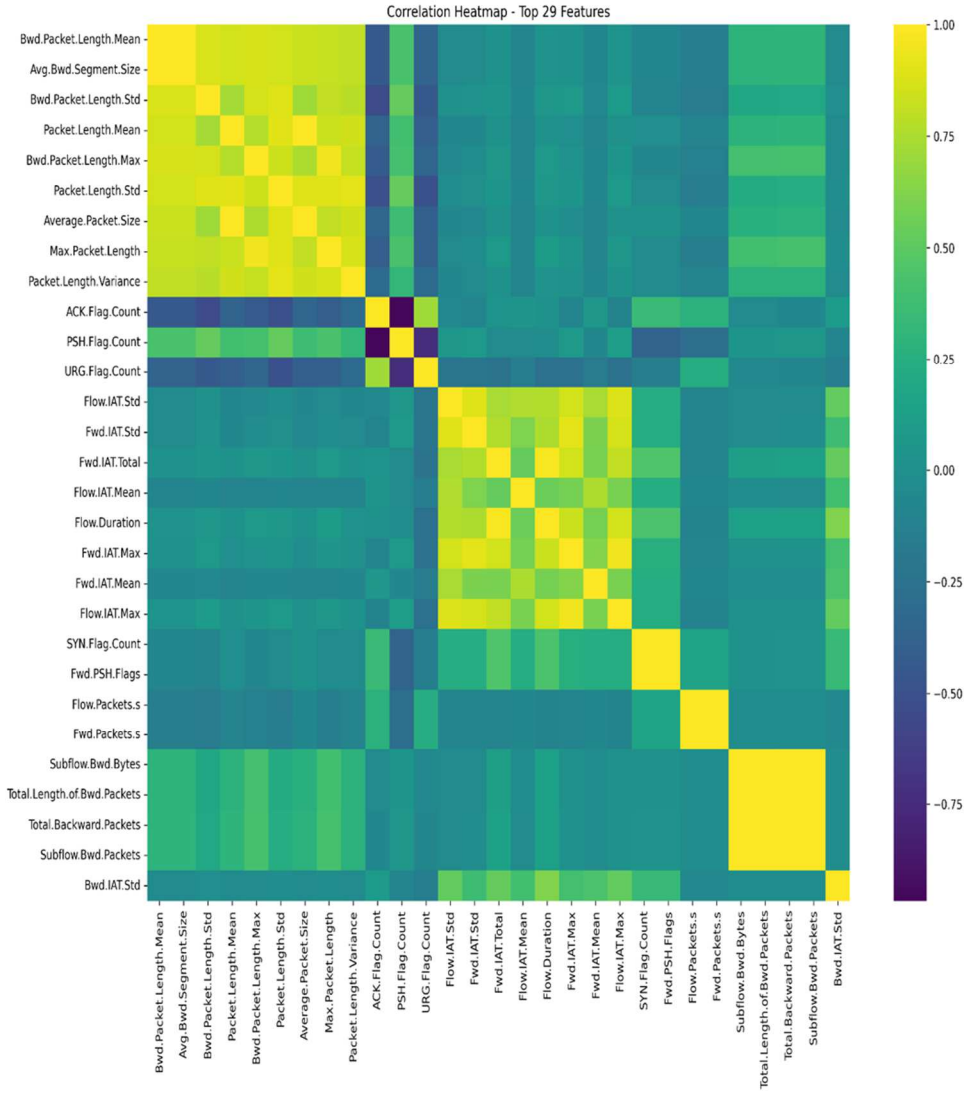


Fig. 9. Correlation heatmap of the 29 most relevant traffic features selected for classification.

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```
Accuracy with Random Forest on Original Dataset 0.9233333333333333

Classification Report:
                precision    recall  f1-score   support

       AMAZON       0.96      0.93      0.94       108
       GOOGLE       0.91      0.91      0.91       105
         HTTP       0.94      0.95      0.95        87
 HTTP_CONNECT       0.80      0.90      0.85        91
   HTTP_PROXY       0.88      0.90      0.89       106
    MICROSOFT       0.99      0.98      0.98        94
          SSL       0.93      0.91      0.92       110
WINDOWS_UPDATE       0.99      0.91      0.95        97
      YOUTUBE       0.91      0.93      0.92       102

     accuracy                           0.92       900
    macro avg       0.93      0.92      0.92       900
 weighted avg       0.93      0.92      0.92       900
```

Fig. 10. RF classification report obtained on the original dataset.

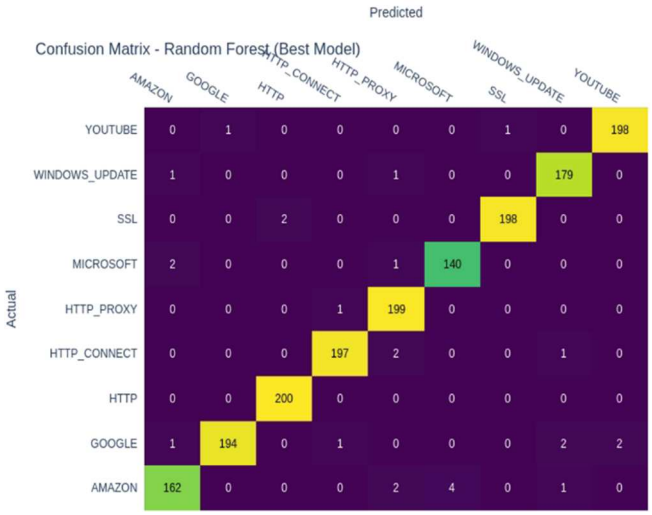


Fig. 11. Confusion matrix obtained using the RF classifier.

After applying feature optimization and parameter tuning procedures, the RF model achieved a significantly improved performance (Fig. 12), reaching an overall classification accuracy of 98.46%. Furthermore, the weighted F1-score exceeded 0.98, demonstrating the effectiveness of the selected feature subset and confirming the suitability of ensemble-learning methods for VPN traffic analysis and application identification.

```
omparing with other algorithms:
andom Forest Accuracy: 0.9846
ogistic Regression Accuracy: 0.8157

est model: Random Forest with accuracy: 0.9846

andom Forest Classification Report:
              precision    recall  f1-score      support
AMAZON         0.975904  0.958580  0.967164   169.000000
OOGLE          0.994872  0.970000  0.982278   200.000000
TTP            0.990099  1.000000  0.995025   200.000000
TTP_CONNECT    0.989950  0.985000  0.987469   200.000000
TTP_PROXY      0.970732  0.995000  0.982716   200.000000
ICROSOFT       0.972222  0.979021  0.975610   143.000000
SL             0.994975  0.990000  0.992481   200.000000
INDOWS_UPDATE  0.978142  0.988950  0.983516   181.000000
OUTUBE         0.990000  0.990000  0.990000   200.000000
ccuracy        0.984643  0.984643  0.984643     0.984643
acro avg       0.984099  0.984061  0.984029  1693.000000
eighted avg    0.984716  0.984643  0.984627  1693.000000

============================================================
INAL RESULTS SUMMARY
```

Fig. 12. Improved RF classification performance after feature optimization.

The SVM classifier was also evaluated using the same dataset and feature set. The obtained confusion matrix, shown in Fig. 13, indicates that the model successfully identified most traffic categories. Nevertheless, a higher number of classification errors can be observed when compared with the Random Forest approach, particularly for traffic classes exhibiting similar statistical characteristics.

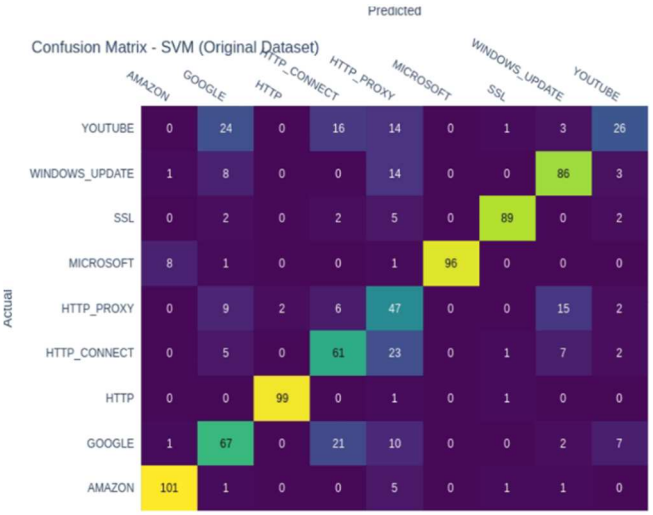


Fig. 13. Confusion matrix obtained using the SVM classifier.

To investigate the feasibility of edge-based deployment, a lightweight neural network model was implemented and converted to TensorFlow Lite format. During training, both training and validation accuracy increased steadily over successive epochs, demonstrating successful convergence of the learning process. The corresponding training evolution is illustrated in Fig. 14.

```
563/563 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.8408 - loss: 0.4849 - val_accuracy: 0.8605 - val_loss:
Epoch 6/10
563/563 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - accuracy: 0.8550 - loss: 0.4539 - val_accuracy: 0.8692 - val_loss:
Epoch 7/10
563/563 ━━━━━━━━━━━━━━━━━━━━ 4s 6ms/step - accuracy: 0.8658 - loss: 0.4105 - val_accuracy: 0.8900 - val_loss:
Epoch 8/10
563/563 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.8802 - loss: 0.3618 - val_accuracy: 0.8917 - val_loss:
Epoch 9/10
563/563 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - accuracy: 0.8840 - loss: 0.3564 - val_accuracy: 0.9053 - val_loss:
Epoch 10/10
563/563 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.8945 - loss: 0.3224 - val_accuracy: 0.9038 - val_loss:
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9144 - loss: 0.2633
[TensorFlow] Accuracy: 90.98%
313/313 ━━━━━━━━━━━━━━━━━━━━ 0s 1ms/step
[TensorFlow] F1-score: 0.8958

================================================
CLASSIFICATION REPORT
================================================
              precision    recall  f1-score   support

      AMAZON       0.85      0.99      0.91       166
 APPLE_ICLOUD       0.00      0.00      0.00         3
 APPLE_ITUNES       0.00      0.00      0.00         3
   CLOUDFLARE       0.90      0.43      0.58        21
```

Fig. 14. TensorFlow Lite training and validation accuracy evolution during model training.

The TensorFlow Lite model achieved a final classification accuracy of 90.98% and an F1-score of 0.8958. Although its classification performance was slightly lower than that obtained by the Random Forest classifier, the model provides an attractive solution for embedded and edge-computing environments due to its reduced computational requirements and efficient inference capabilities.

Finally, the developed framework was validated in a real-time monitoring scenario executed directly on the Raspberry Pi platform. The system successfully identified and classified multiple traffic categories, including SSL, GOOGLE,

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YOUTUBE, and HTTP_PROXY flows during live network operation. An example of the real-time traffic classification results is presented in Fig. 15, confirming the capability of the proposed framework to perform intelligent traffic monitoring under practical operating conditions.

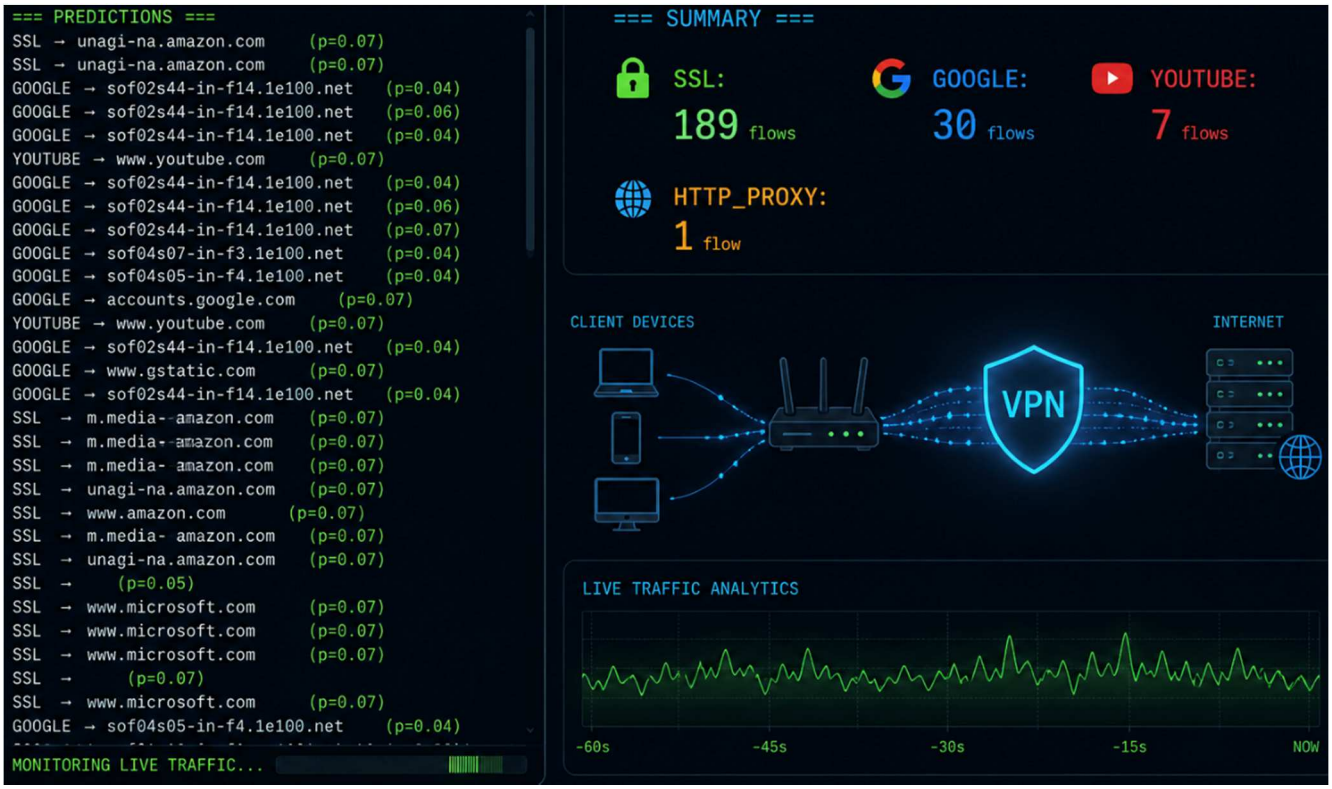


Fig. 15. Real-time traffic classification results obtained during live network monitoring.

The obtained results demonstrate that ML techniques can effectively classify encrypted and application-level network traffic using statistical flow characteristics without requiring access to packet payload contents. Among the evaluated approaches, the Random Forest classifier achieved the highest overall performance, while the TensorFlow Lite implementation provided an effective compromise between classification accuracy and deployment feasibility on resource-constrained edge devices.

V. Conclusion

This paper presented an intelligent VPN traffic analysis and classification framework implemented on a Raspberry Pi edge-computing platform using ML techniques. The proposed solution integrates secure VPN communication, real-time traffic monitoring, feature extraction, and intelligent traffic classification within a unified architecture designed for resource-constrained environments. By combining OpenVPN and WireGuard services with ML-based analysis, the framework enables enhanced visibility into encrypted network traffic without requiring access to packet payload contents.

Three ML approaches were evaluated, namely RF, SVM, and TensorFlow Lite-based neural network inference. Experimental results demonstrated that all evaluated models could identify and classify network traffic flows with satisfactory performance. Among the tested approaches, the RF classifier achieved the best overall performance, reaching a classification accuracy of 98.46% and a weighted F1-score exceeding 0.98. The TensorFlow Lite implementation achieved an accuracy of 90.98%, demonstrating that efficient traffic classification can also be performed directly on embedded edge devices with limited computational resources.

The obtained results confirm the effectiveness of statistical traffic features for identifying application-level communication patterns, even when traffic is protected by VPN encryption mechanisms. Furthermore, the successful deployment of the classification framework on Raspberry Pi demonstrates the feasibility of implementing intelligent cybersecurity services on low-cost hardware platforms suitable for SOHO environments and edge-computing scenarios.

Future research directions include the investigation of DL architectures specifically designed for encrypted traffic analysis, the integration of anomaly and intrusion detection mechanisms, and the evaluation of the proposed framework under larger-scale network environments. Additional work will also focus on improving real-time inference performance and extending the system to support adaptive and continuously updated ML models capable of responding to emerging network threats and evolving traffic patterns.

The proposed framework demonstrates that the combination of VPN technologies, edge computing, and ML can provide an effective and practical solution for intelligent network monitoring, contributing to the development of lightweight cybersecurity systems capable of operating efficiently in modern distributed networking environments.

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